# Jamil Hawwash Data Scientist

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AI Engineer and Data Scientist specializing in production-grade AI systems, including multilingual voice agents, RAG pipelines, and intelligent automation. Experienced in designing and deploying end-to-end AI solutions that reduce manual workflows, improve decision-making, and scale real-world applications.

## Education

**BSc. Data Science & Artificial Intelligence** 09/2020 – 07/2024 | Amman, Jordan
Princess Summaya University for Technology

## Technical Skills

**A.I & ML:** PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, Qdrant, Haystack, Pipecat, Prompt Engineering

**Data Engineering & Analysis:** Excel, Power BI, Matplotlib, Seaborn, PySpark, Pandas, NumPy

**Programming:** Python, SQL, R, C++

**Automation & Tools:** Power Automate, Selenium, n8n, Git, FastAPI, AWS (S3, SES, Sagemaker), Postgres, RabbitMQ, Docker

## Professional Experience

**A.I Engineer, Egnite** 06/2025 – Present | Amman, Jordan
- Developed and deployed end-to-end multilingual RAG-based AI voice agents using Pipecat, Haystack, Qdrant, and Gemini/OpenAI Realtime, leveraging function calling to automate scheduling, ticketing, and support workflows, reducing manual agent handling by ~35% for enterprise clients.
- Containerized and deployed real-time speech pipelines, training custom XTTS voice models using Docker on RunPod and AWS, achieving sub-500 ms transcription latency and production-grade Arabic speech synthesis.
- Built and scaled backend microservices for ticketing, scheduling, and notifications using Python, SQLAlchemy, FastAPI, and AWS (S3, SES), supporting 5k+ automated interactions per month with high service reliability.
- Optimized prompt design and multilingual dialogue flow across Jordanian, Iraqi, and Saudi Arabic dialects, improving contextual accuracy and user satisfaction by ~25% while reducing fallback rates by ~30%
- Implemented a system-wide health monitoring and alerting framework for microservices and external AI providers (OpenAI Realtime, ElevenLabs, Azure), enabling rapid failover and preventing customer-facing service disruptions via automated email and WhatsApp alerts.

**Data Science Intern, Miral Destinations** 02/2025 – 06/2025 | Abu Dhabi, UAE
- Analyzed user-bot interactions using Python to identify behavioral patterns, reducing bot-to-agent transfer rates by 30% and improving engagement quality.
- Developed a Python-based NLP system with regex and token metrics to categorize chatbot queries, revealing top FAQs and failure points.
- Automated weekly reporting and testing using Python, Excel, and Power BI to track FAQs, failure points, and agent handoff triggers.
- Collaborated with the CRM team to perform revenue analysis across all tenants, uncovering sales trends and supporting strategic decision-making for tenant management.

**Data Analysis Intern, Jaramed** 08/2023 – 05/2024 | Amman, Jordan
- Developed an end-to-end ETL pipeline using Python and SQL to extract, clean, and structure pharmaceutical sales data, improving data quality and analysis readiness.
- Performed demand forecasting and sales analysis using different machine learning models (XGBoost, LSTMs, ARIMA) to uncover seasonal trends and optimize distribution strategies across multiple countries.

## Projects

**Real-Time Driver Drowsiness Detection**
- Developed a CNN model with a robust architecture using TensorFlow to detect drowsiness from facial cues.
- Trained the model on our own dataset, consisting of 2,000 images from volunteers, achieving an accuracy of over 95%. Used techniques like data augmentation to decrease overfitting.
- Deployed the model on Raspberry Pi with a camera and buzzer connected for real-time detection and alerts.

**CNN-based COVID-19 X-ray Classifier**
- Trained a CNN from scratch to classify chest x-ray images as COVID-19 or normal.
- Applied data augmentation and preprocessing for model robustness, achieving a 98% accuracy in classifying COVID-19 cases from X-rays.

## Certificates and Awards

**1st Place Hackathon Winner - AWS Innov8 Bahrain**
- Co-developed an AI-powered medical chatbot for OSCE exam simulation; earned 1st place out of 20 teams for innovation and real-world applicability.

**Change Management - TAAEEN**

## Additional Information

- UAE Golden Visa Holder
- Driving License

## Languages

- English
- Arabic